Capstone Progress Update 1: Weeks 1 and 2

An Automated Data Pipeline for the German Electricity Market

1. Where I stand

After two weeks the foundation of my project is in place. In the first week I checked that I can get all the data I need, easily and for free, and I set up the project. In the second week I built the first part of the pipeline: the scripts that download the data and store it unchanged in a database, with every run written to a log. I am on track with the plan. The cleaning and the analysis come next.

2. What I did

Week	What I did	How it went
1	Tested the electricity price and generation data from Energy-Charts	Both worked without an account or key, with a value every 15 minutes
1	Looked at the German weather service's hourly temperature files	The folder is online and most files were updated the same day
1	Set up the project, the database and a few automatic tests	It runs, and the tests pass
2	Built the download scripts for prices, generation data and weather	They retry when a request fails
2	Stored everything unchanged in raw tables, with source and load time	Raw data can always be rebuilt from the original download
2	Added a run log that records every run and its result	This is the base for checking how reliable the pipeline is

3. What I found about the data

3.1 Electricity prices

The day-ahead prices for Germany and Luxembourg come in euros per megawatt hour, one value every 15 minutes. They come from the Bundesnetzagentur's SMARD platform, so I have a second route to the same data if Energy-Charts is ever down.

3.2 Power generation and renewable share

The generation data has 21 series, also every 15 minutes. It covers solar, wind, biomass, hydropower, coal, gas, total load and residual load, and the share of renewables. That is good news, because I do not have to calculate the renewable share myself. The response does not say which unit the generation numbers use. The size of the wind values suggests megawatts, and I will confirm this in the documentation before I rely on it.

3.3 Weather

The German weather service offers one file per station with hourly temperatures. Four stations had not been updated for months, so I will leave them out. Weather is hourly, while prices and generation come every 15 minutes, so I will bring them to the same time step in the cleaning stage.

3.4 What I still have to check

My first tests used short time windows. I still need to confirm how far back the history goes and how the services behave with many requests in a row. The SMARD downloads are my backup source, and I have not looked at their file formats yet.

4. How the pipeline works so far

Step	What happens
1. Download	Scripts ask the services for prices, generation data and weather, and try again if a request fails
2. Raw tables	Everything is stored exactly as downloaded, with its source and the time of loading
3. Run log	Each run records what was loaded, how many rows, and whether it worked
4. Next	Cleaning and quality checks, then the analysis tables

5. What this means for my questions

My question	Can I answer it with this data?
1. How do prices change by hour, weekday and season?	Yes, once enough history is loaded
2. How much does the share of renewables move the price?	Yes. The renewable share comes ready-made, and weather comes from the weather service
3. When are the cheapest and greenest hours?	Yes, with prices and renewable share
4. Can I forecast tomorrow's prices?	Probably yes, but it needs a long history
5. Does my pipeline run reliably?	The run log is in place and fills up with every run

6. What could go wrong

What could go wrong	What I do about it
The unit of the generation data is not stated	I check it in the service documentation before using the numbers
A service could be down on some days	Retries, the run log, and the SMARD download as a backup
Different time steps could break joins	I convert everything to the same step in the cleaning stage and test for gaps and duplicates

Sources

Energy-Charts: api.energy-charts.info. SMARD, Bundesnetzagentur: smard.de. Deutscher Wetterdienst open data: opendata.dwd.de.